

Factors impacting effective altruism

Revisiting heuristics and biases in charity in a replication and extensions
Registered Report of Baron and Szymanska (2011)

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3 NOVEMBER 2023

Background

- Some people donate money to charities
- Presumably, the point of doing this is to help people
- Do people really donate just to help?
- Or are they affected by other considerations?

Why do people donate?

- Bekkers and Wiepking (2011)
- Very influential paper (>1900 citations and counting)
- Categorises the reasons why individual people donate
 - **(a) awareness of need** Are they aware that people need help?
 - **(b) solicitation** Were they asked to donate?
 - **(c) costs and benefits** How do the benefits compare with the cost?
 - **(d) altruism** Does donating help the beneficiaries?
 - **(e) reputation** Can donating affect their social status?
 - **(f) psychological benefits** Does donating make them feel happy?
 - **(g) values** Does helping the beneficiaries fit their values?
 - **(h) efficacy** Do they feel like donating makes a difference?

Basic design

- Baron and Szymanska (2011)
- Utilitarian model: “economic” model for what the “best” way to donate is
 - **Cost-effectiveness:** maximise the amount of Good that is done per X amount of money
- If two charities, A and B, do the same amount of Good per X amount of money, there is no difference in cost-effectiveness between the two
- If you’re still donate more towards charity A over charity B, then there must be a bias in play
- How do we study these biases? -> Replication and extensions of a previous study

Replications?

- “Replication crisis” in science – many studies are found not to “work” when tried again (Klein et al., 2015)
- Prevalent in social psychology
 - Related: Dan Ariely and Francesca Gino “honesty nudging” scandal
- Movement to try and replicate popular studies to verify them
- Most often with a greater sample size
- Extensions: additional hypotheses tacked on to the side of the replication

Basic design

- Study design:
- We have two charities, Charity A and Charity B
- Both have an **equal cost-effectiveness**
 - i.e. On average, Charity A and Charity B do the same amount of good for the same amount of money the participant gives to them
- The only difference between the two is [insert manipulation here]
- Given that you have X amount of money to donate, how much money will you donate to each charity?

Basic design

- Main statistical test: **one-sample t-test** (to see if the distribution is different from max cost-effectiveness, e.g. 50% to each charity)
- Not just one version per hypothesis
- Multiple items presented -> validity over different framings?
- ANOVA and paired t-tests

How much would you allocate to A/B?

A:100%, B:0%	A:90%, B:10%	A:80%, B:20%	A:70%, B:30%	A:60%, B:40%	A:50%, B:50%	A:40%, B:60%	A:30%, B:70%	A:20%, B:80%	A:10%, B:90%	A:0%, B:100%
☐	☐	☐	☐	☐	☒	☐	☐	☐	☐	☐

This study is about contributions to worthy causes, such as helping people in need who cannot help themselves, people such as young children or those who are ill. Imagine that you are not one of these needy people and that, in fact, you have enough money so that it is easy for you to give some away for such causes without seriously hurting your quality of life. Imagine that you are willing to contribute some of your annual income to such causes.

Unless otherwise specified, suppose that all programs mentioned are equally effective and efficient.

Instructions to participants

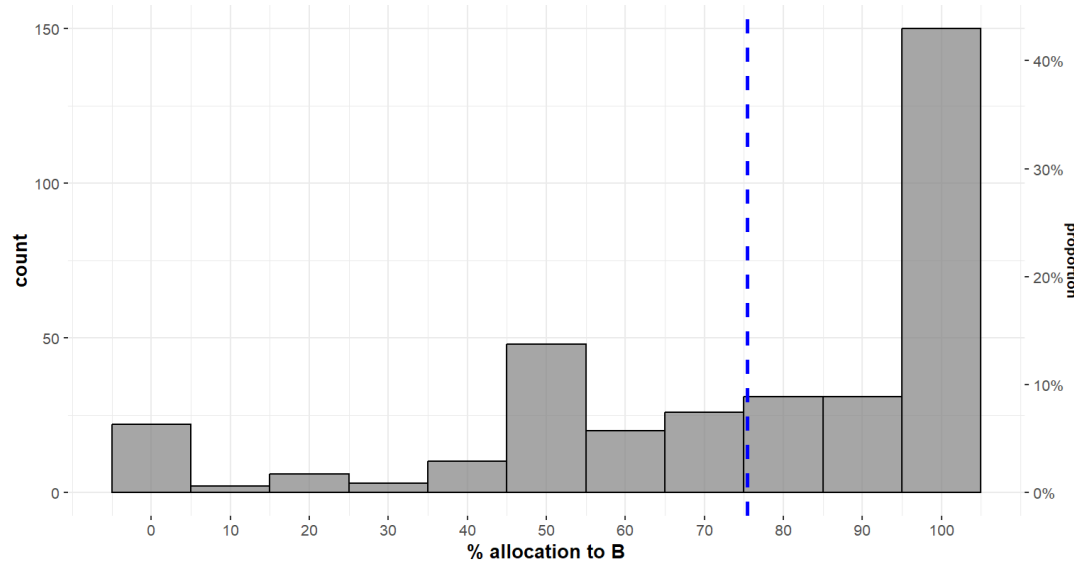
Replication hypothesis 1: Perceived waste

- Do charities perceived to be wasteful affect donation allocation?
- Example item:

“A and B [...] prevent 5 deaths for every \$1,000 of donations.
A spends \$200 out of every \$1,000 of donations **on advertising.**
B spends \$100.”

H1 (Waste) - Study 1

$t_{\text{Student}}(348) = 16.03$, $p = 1.06\text{e-}43$, $\hat{g}_{\text{Hedges}} = 0.86$, $\text{CI}_{95\%} [0.73, 0.98]$, $n_{\text{obs}} = 349$



$\log_e(\text{BF}_{01}) = -92.79$, $\hat{\delta}_{\text{posterior difference}} = 25.30$, $\text{CI}_{95\%}^{\text{ETI}} [22.27, 28.43]$, $r_{\text{JZS Cauchy}}^{\text{JZS}} = 0.71$

Replication hypothesis 1: Perceived waste

- Do charities perceived to be wasteful affect donation allocation?
- Very much yes. (M = 75.42% to B, $p < .001$, $d = 0.86$)

Replication hypothesis 2: Past costs

- Do charities that have high past costs affect donation allocation?
- Example item:

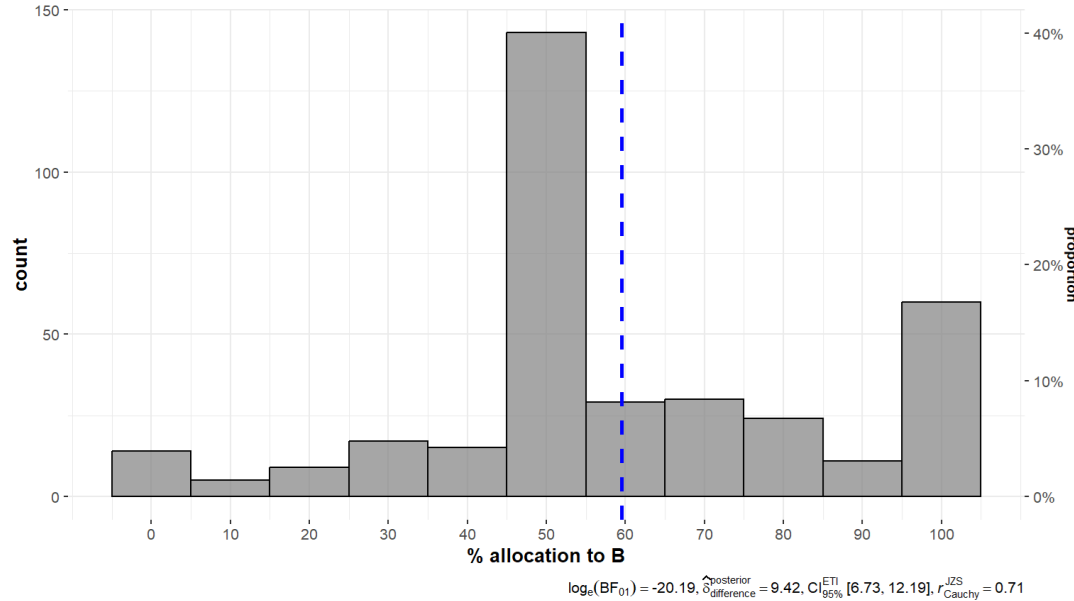
“A and B will each prevent 5 deaths for every \$10,000 of new donations.

A was much more expensive to get started.

Thus, the cost per life saved on the average is higher for A, because A has spent more money in total.”

H2 (Average cost) - Study 2

$t_{\text{Student}}(356) = 7.04, p = 1.00\text{e-}11, \hat{g}_{\text{Hedges}} = 0.37, \text{CI}_{95\%} [0.26, 0.48], n_{\text{obs}} = 357$



Replication hypothesis 2: Past costs

- Do charities that have high past costs affect donation allocation?
- Also yes. (M = 59.5% to B, $p < .001$, $d = 0.37$)

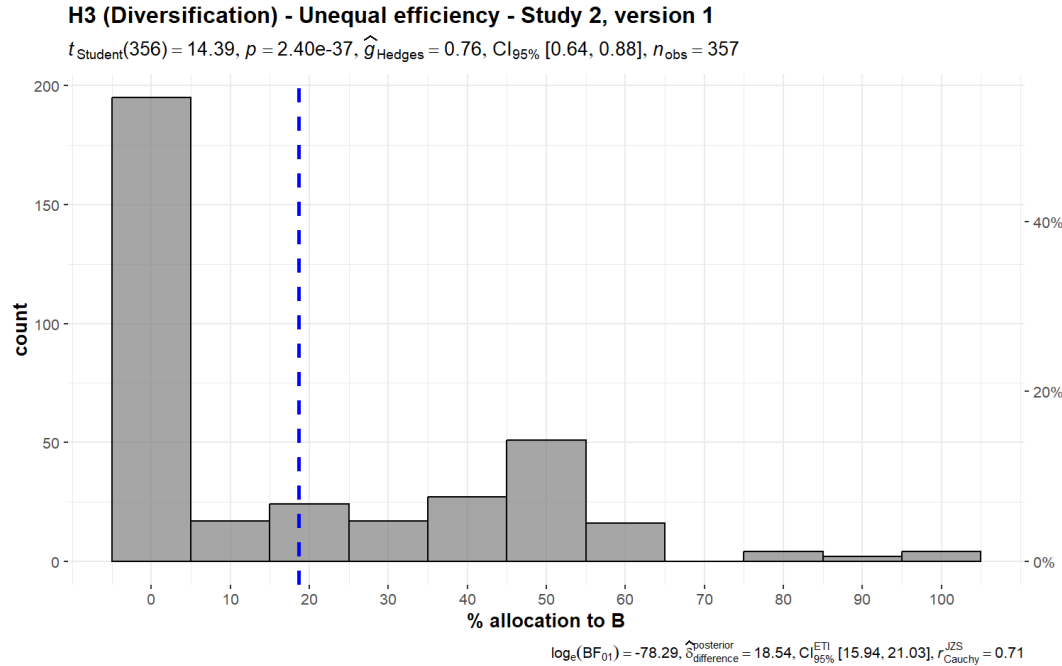
Replication hypothesis 3: Diversification

- Would participants donate more than nothing to a less effective charity?
- Example item:

A can save **one life for \$10,000**. B can save **one life for \$12,500**.

The people helped are from the same groups, with the same problems.

Replication hypothesis 3: Diversification



- Would participants donate more than nothing to a less effective charity?
- Still yes. (M = 18.63% to B, $p < .001, d = 0.76$)

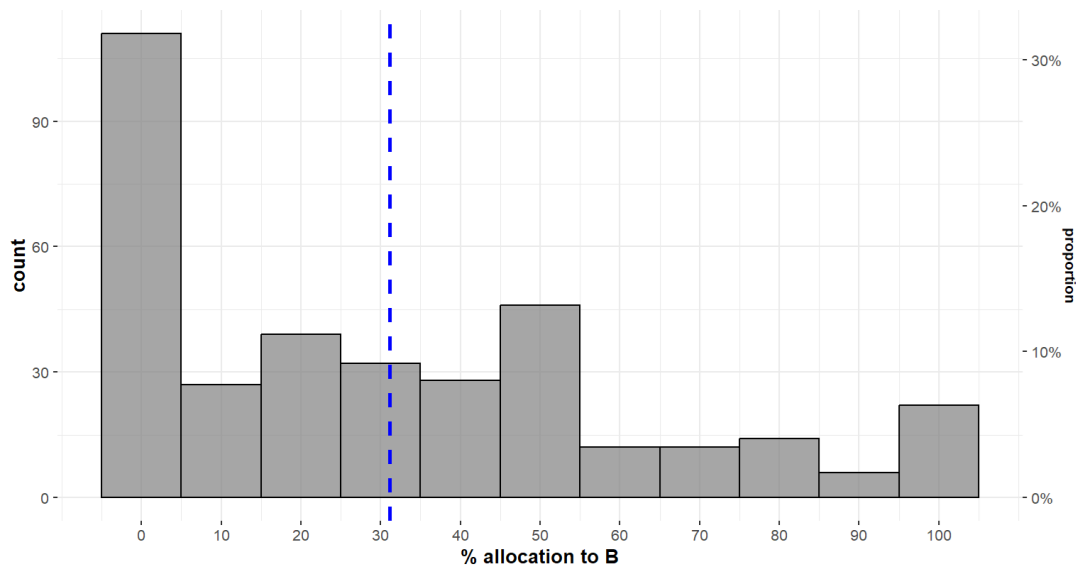
Replication hypothesis 3: Diversification

- How about when the less effective charity runs several projects?
- Example item:

“A puts all the money into **one project**,
which has a 75% chance of helping many children [...]
B puts the money into **several different projects**,
each of which has a 70% chance of helping some children [...].”

H3 (Diversification) - Unequal efficiency, several projects versus one - Study 1

$t_{\text{Student}}(348) = 18.87, p = 3.47\text{e-}55, \hat{g}_{\text{Hedges}} = 1.01, \text{CI}_{95\%} [0.88, 1.14], n_{\text{obs}} = 349$



$\log_e(\text{BF}_{01}) = -119.02, \hat{\sigma}_{\text{difference}}^{\text{posterior}} = 31.01, \text{CI}_{95\%}^{\text{ETI}} [27.64, 34.27], r_{\text{JZS Cauchy}}^{\text{JZS}} = 0.71$

Replication hypothesis 3: Diversification

- How about when the less effective charity runs several projects?
- In fact this makes them donate even more. ($M = 31.09\%$ to B, $p < .001$, $d = 1.01$)

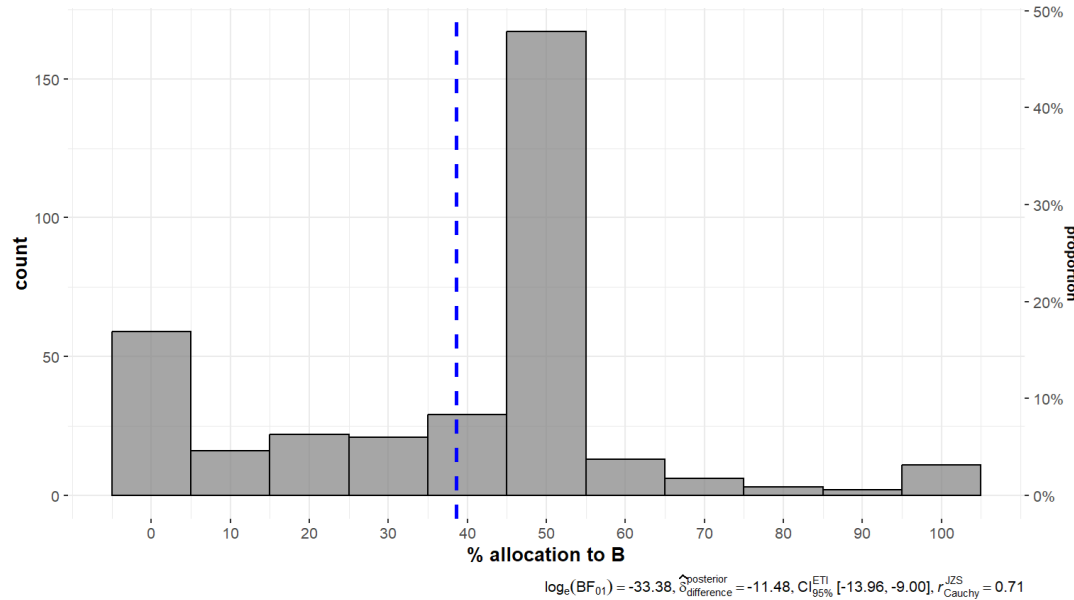
Replication hypothesis 4: Ingroup effect

- Do participants allocate more money to their ingroup?
- Example item:

“A helps children who are in **your own country**.
B helps children **around the world**.
The children are equally needy.”

H4 (Nationalism) - Study 1

$t_{\text{Student}}(348) = -9.01, p = 1.42\text{e-}17, \hat{g}_{\text{Hedges}} = -0.48, \text{CI}_{95\%} [-0.59, -0.37], n_{\text{obs}} = 349$

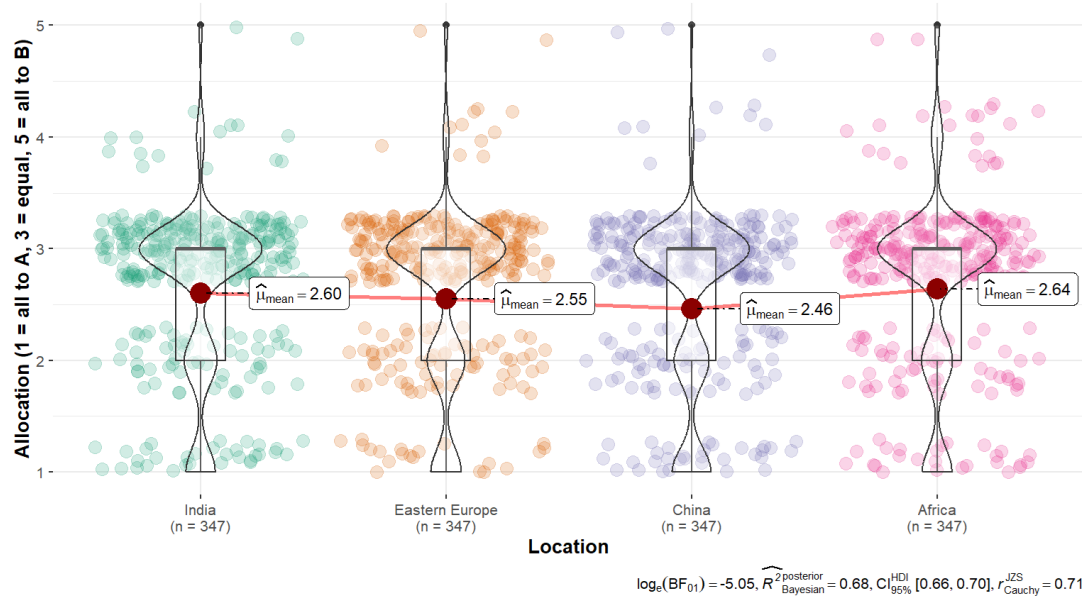


Replication hypothesis 4: Ingroup effect

- Do participants allocate more money to their ingroup?
- Obviously. (M = 38.57% to B, $p < .001, d = -0.48$)

H4 (Nationalism) - Study 3

$F_{\text{Fisher}}(2.94, 1019) = 8.49, p = 1.66\text{e-}05, = 5.15\text{e-}03, \text{CI}_{95\%} [0.00, 1.00], n_{\text{pairs}} = 347$



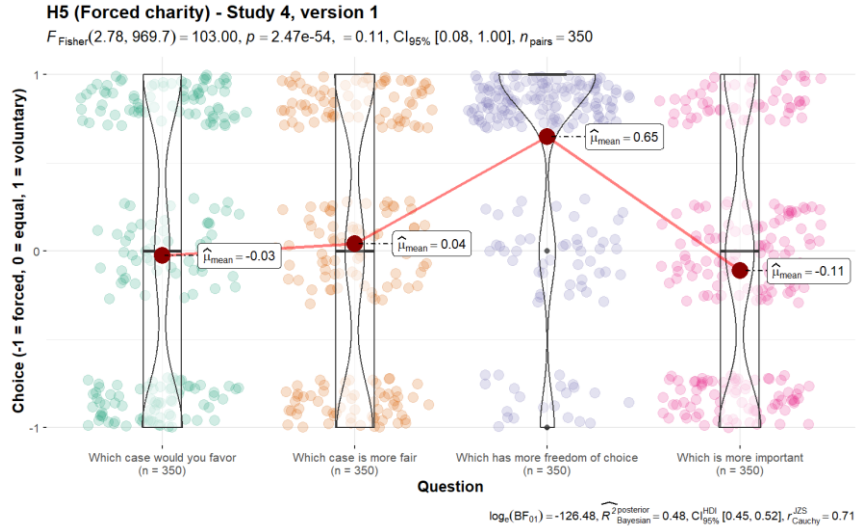
Replication hypothesis 4: Ingroup effect

- Another version: Participants were given a location (“[... Charity] B cures the same disease in children in X location.”)
- Very slight differences depending on location ($p < .001$, partial $\omega^2 = 0.00$)

Replication hypothesis 5: “Forced charity”

- If both a **tax (“forced charity”)** and a **charity** can help others, then would participants prefer donations over tax?
- Five different sets of items were presented
- Slightly different setup: responses over four questions were averaged
 - “Which case would you favor?”
 - “Which case is more fair?”
 - “Which has more freedom of choice?”
 - “Which [of fair cost allocation and freedom of choice] is more important?”

Replication hypothesis 5: “Forced charity”



- Nominally replicated: All five items were $p < .001$
- ANOVA tests found that one question stood out: the “which has more freedom of choice” question
 - Perhaps it should have been a comprehension check
- After this question was removed, 4/5 versions suddenly became not significant

Replication hypothesis 5: “Forced charity”

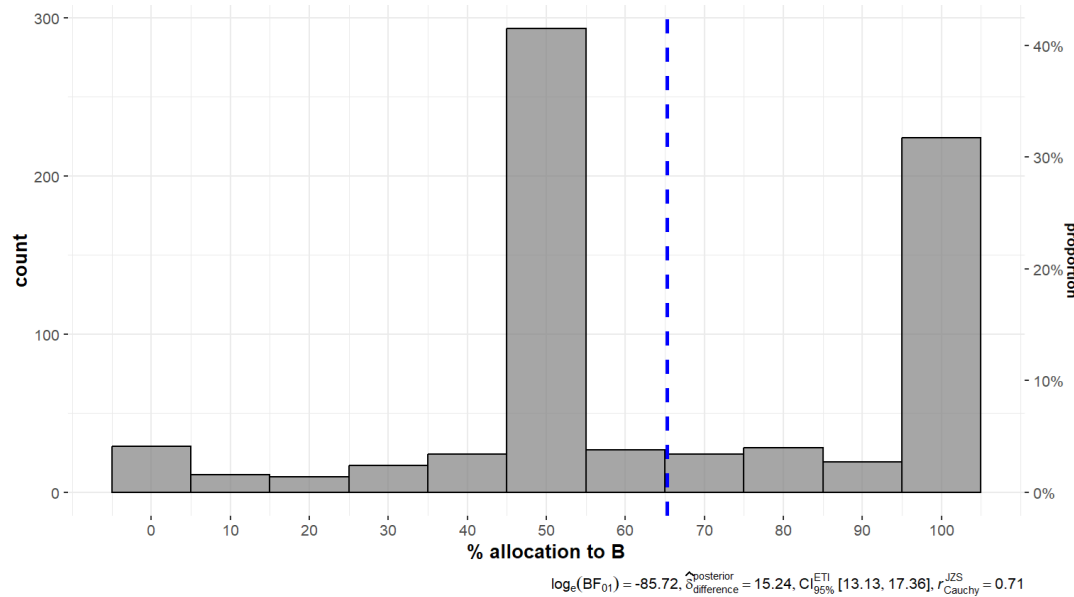
- The effect of the last one could be explained by political factors
 - “A new epidemic disease threatens **to infect 10% and kill 1% of the children** in your nation. It can be cured by a treatment that costs too much [...] Additional funds are needed.” -> choice between income tax increase and donations
 - Participant feedback: “don't think it's fair to raise taxes on everyone, especially when a disease has such a **low [...] death rate**” (?)
- My guess: medical populism (study participants are from the US)
- So does “forced charity” as a bias even exist?

Extension hypothesis 1: Reputation

- Original research question!
- Item posed: “A **publishes the names** of donors and how much they donated on their website. B keeps donors **anonymous**.”
- My hypothesis: Being provided with a means to increase your own reputation would provide an incentive towards choosing A

H6 (Reputation)

$t_{\text{Student}}(705) = 14.25, p = 1.15\text{e-}40, \hat{g}_{\text{Hedges}} = 0.54, \text{CI}_{95\%} [0.46, 0.61], n_{\text{obs}} = 706$



Extension hypothesis 1: Reputation

- Effect found in the opposite direction
- People instead preferred B, the anonymous option
- $M = 65.27\%$ towards B, $p < .001$, $d = 0.54$
- Literature review was mixed-to-negative on anonymity

Extension hypothesis 2: Overhead costs

- Conceptual replication of Gneezy et al. (2014)
- People don't like to pay administrative costs for donations
- What if we tell donors that another donor will pay for the admin costs of your donation?

BEHAVIORAL ECONOMICS

Avoiding overhead aversion in charity

Uri Gneezy,^{1,2*} Elizabeth A. Keenan,¹ Ayelet Gneezy¹

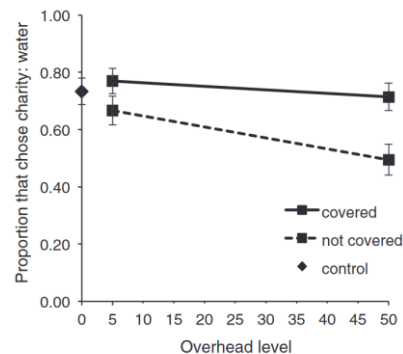


Fig. 1. Proportion of participants that chose charity: water by overhead level and by whether or not someone else covered the overhead. Error bars are ± 1 SEM.

Extension hypothesis 2: Overhead costs

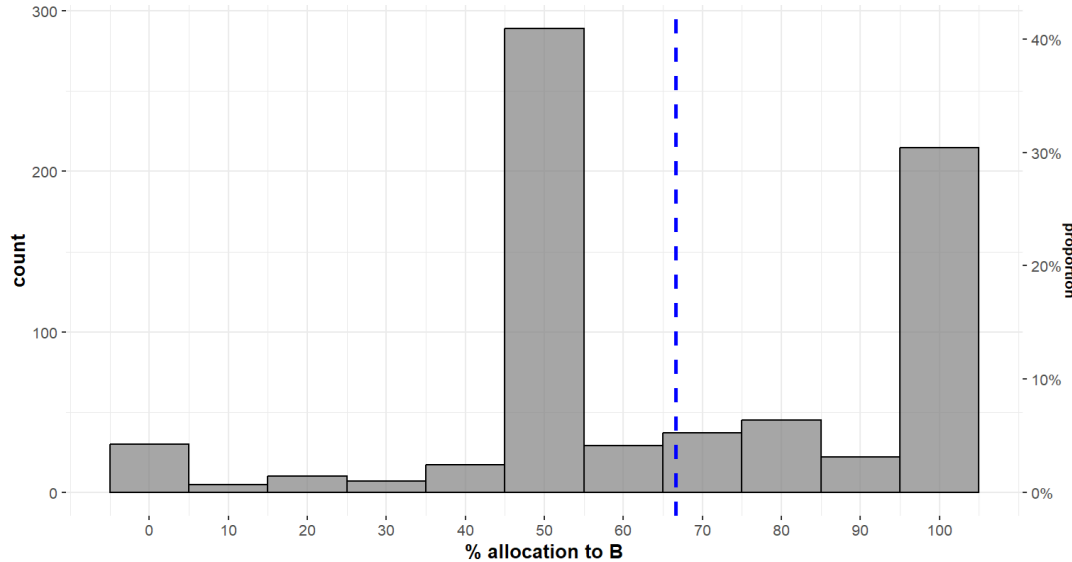
- Item posed:

“A and B [...] spend **50%** of the donations they receive on **administrative costs**. For each \$100 contribution to A, \$50 will go to helping children and \$50 will be used to cover administrative costs.

For each \$100 contribution to B, all \$100 will go to helping children; **another donor will cover the corresponding \$100** administrative cost of this contribution.”

H7 (Overhead cost)

$t_{\text{Student}}(705) = 16.01, p = 2.02\text{e-}49, \hat{g}_{\text{Hedges}} = 0.60, \text{CI}_{95\%} [0.52, 0.68], n_{\text{obs}} = 706$



$\log_e(\text{BF}_{01}) = -105.73, \hat{\delta}_{\text{difference}}^{\text{posterior}} = 16.55, \text{CI}_{95\%}^{\text{ETI}} [14.54, 18.53], r_{\text{Cauchy}}^{\text{JZS}} = 0.71$

Extension hypothesis 2: Overhead costs

- This “trick” works on the participants! (M = 66.57% to B, $p < .001$, $d = 0.60$)

Conclusions and further directions

- Successful replication (5/5), original studies mostly validated
- Extensions: one followed the hypothesis, one went in the other direction
- Small range of biases included
 - Items can easily be swapped out for others
- Very easy to modify it to run between-group versions

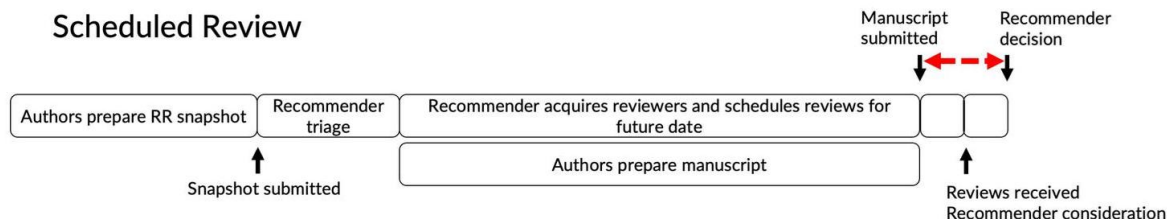
Registered report?

In short: plan -> peer review stage 1 -> actual study -> peer review stage 2

- Submit a full plan of what you are going to do (incl. the analyses you will run), a.k.a. a **preregistration**
- Peer review as necessary
- Study only begins when the plan is accepted; peer review of the final paper happens after

My study:

Scheduled Review



(Two rounds of reviews happened during stage 1 for my paper)

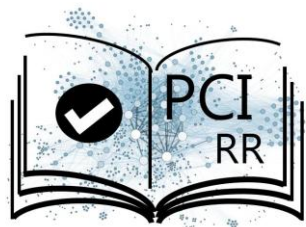
Peer Community In (PCI) Registered Reports

Platform of researchers which provides free (both *libre* and *gratis*) peer reviews for registered reports

After stage 2 peer review, a **recommendation** is given

A list of PCI RR-friendly journals accept the recommendation as peer review and will not:

- “[re-judge] the **importance, novelty or timeliness** of a research question”
 - “relitigate any [already approved] scientific elements of the study”
- as these were judged *before* the study happens



Practical considerations for PCIRR theses

PROS

Learn good research methods

- Reduces the possibility of picking up questionable research methods such as p-hacking and cherry-picking
- Lets others in the field see your work as it happens and recommend improvements

Reduces worry about results

- No need to worry about being rejected by a journal for reasons like not having a significant result

CONS

Higher skill barrier

- Understanding reproducible data analysis code removes a layer of abstraction
- Not all students may be able/willing to commit to such rigour

Tighter deadlines

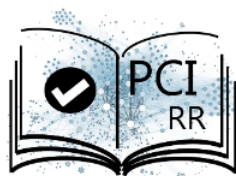
- Dependant on outside peer review so deadlines are much more inflexible

Incorporating open research practices into the undergraduate curriculum increases understanding of such practices

Corina Logan based on reviews by Kelsey McCune, Neil Lewis, Jr., Lisa Spitzer and 1 anonymous reviewer

A recommendation of:

STAGE 2



Evaluating the Pedagogical Effectiveness of Study Preregistration in the Undergraduate Dissertation

Madeleine Pownall, Charlotte R. Pennington, Emma Norris, Marie Juanchich, David Smalle, Sophie Russell, Debbie Gooch, Thomas Rhys Evans, Sofia Persson, Matthew HC Mak, Loukia Tzavella, Rebecca Monk, Thomas Gough, Christopher SY Benwell, Mahmoud Elsherif, Emily Farran, Thomas Gallagher-Mitchell, Luke T. Kendrick, Julia Bahnmüller, Emily Nordmann, Mirela Zaneva, Katie Gilligan-Lee0, Marina Bazhydai, Andrew Jones, Jemma Sedgmond, Iris Holzleitner, James Reynolds, Jo Moss, Daniel Farrelly, Adam J. Parker, and Kait Clark

<https://psyarxiv.com/xg2ah>

version 1

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Pownall and colleagues found that preregistration did not affect attitudes toward QRPs, but it did improve student understanding of open research practices. Using exploratory analyses, they additionally found that those who preregistered were those students who reported that they had more opportunity, motivation, and greater capability. This shows how important it is to incorporate the teaching of open research practices such that students can increase their capability, motivation, and opportunity to pursue such practices, whether it is preregistration or other practices that are better known to reduce QRPs (such as registered reports; Krypotos et al. 2022).